

Software architecture

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Software development as model transformations

The Robustness Analysis process

Lecture: The Robustness Analysis process

Module: From requirements model to design model

The Robustness Analysis Process

The Robustness Analysis process in 4 steps

1. Select a use case scenario (or activity diagram).
2. Place the actor in the UML Collaboration diagram.
3. For every step *“The actor does...”* in the use case scenario:
 - a. Identify and add Boundary, Control, and Entity components.
 - b. Draw the associations between these components.
 - c. Label the actions performed by each component to satisfy the interactions in the use case.
4. Convert the Collaboration diagram into a Sequence diagram (optional)

Step 1 — Select a use case scenario

- It can be the **main flow** or an **alternate flow**.
- We **selected** the **main flow** of the scenario *"Create a guest"*.
 - ✓ It could be a **scenario** from a **use case** named **Manage Guests**.
 - ✓ It could be a **scenario** **associated** with a **user story**, named *"Create Guest"*.
 - ✓ ...

Use case scenario: Create a Guest

Pre-condition(s):

- A screen with an option *"Create new Guest"* is visible.
- The Reservation Manager is authenticated.

Main flow:

- 1.The Reservation Manager selects the option *"Create new Guest"*.
- 2.The system shows a form requesting Guest data.
- 3.The Reservation Manager enters the form data.
- 4.The systems shows a message *"Guest created sucessfully"*.

Alternate flow(s):

....
....

Step 2: Place the actor in the diagram

Use case scenario: Create a Guest

Pre-condition(s):

- A screen with an option "*Create new Guest*" is visible.
- The Reservation Manager is authenticated.

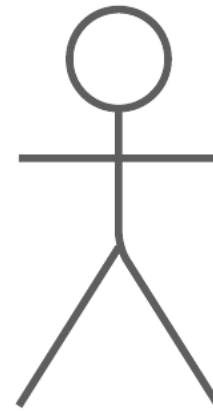
Main flow:

- 1.The Reservation Manager selects the option "*Create new Guest*".
- 2.The system shows a form requesting Guest data.
- 3.The Reservation Manager enters the form data.
- 4.The systems shows a message "*Guest created sucessfully*".

Alternate flow(s):

....

....



**Reservation
Manager**

Step 3 - For every step *“The actor does..”*

Use case scenario: Create a Guest

Pre-condition(s):

- A screen with an option *“Create new Guest”* is visible.
- The Reservation Manager is authenticated.

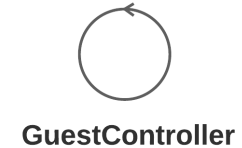
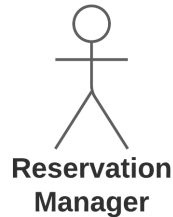
Main flow:

1. The Reservation Manager selects the option *“Create new Guest”*.
2. The system shows a form requesting Guest data.
3. The Reservation Manager enters the form data.
4. The systems shows a message *“Guest created sucessfully”*.

Alternate flow(s):

....

....



Step 3a - Identify Boundary, Control, and Entity components

Step 3 - For every step *"The actor does.."*

Use case scenario: Create a Guest

Pre-condition(s):

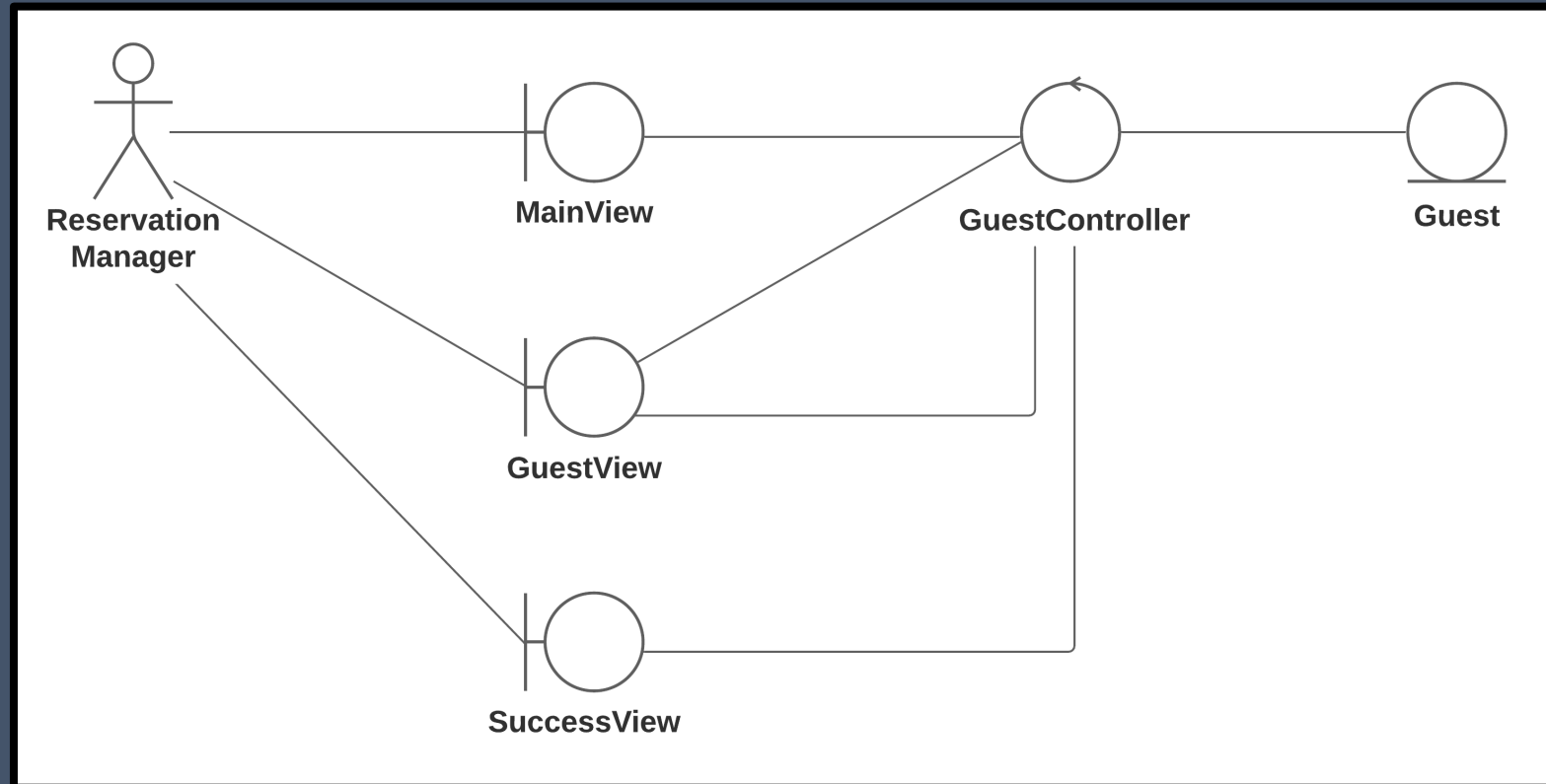
- A screen with an option "Create new Guest" is visible.
- The Reservation Manager is authenticated.

Main flow:

1. The Reservation Manager selects the option "Create new Guest".
2. The system shows a form requesting Guest data.
3. The Reservation Manager enters the form data.
4. The systems shows a message "Guest created sucessfully".

Alternate flow(s):

....
....



Step 3b – Draw the associations between the components

Step 3 - For every step *"The actor does.."*

Use case scenario: Create a Guest

Pre-condition(s):

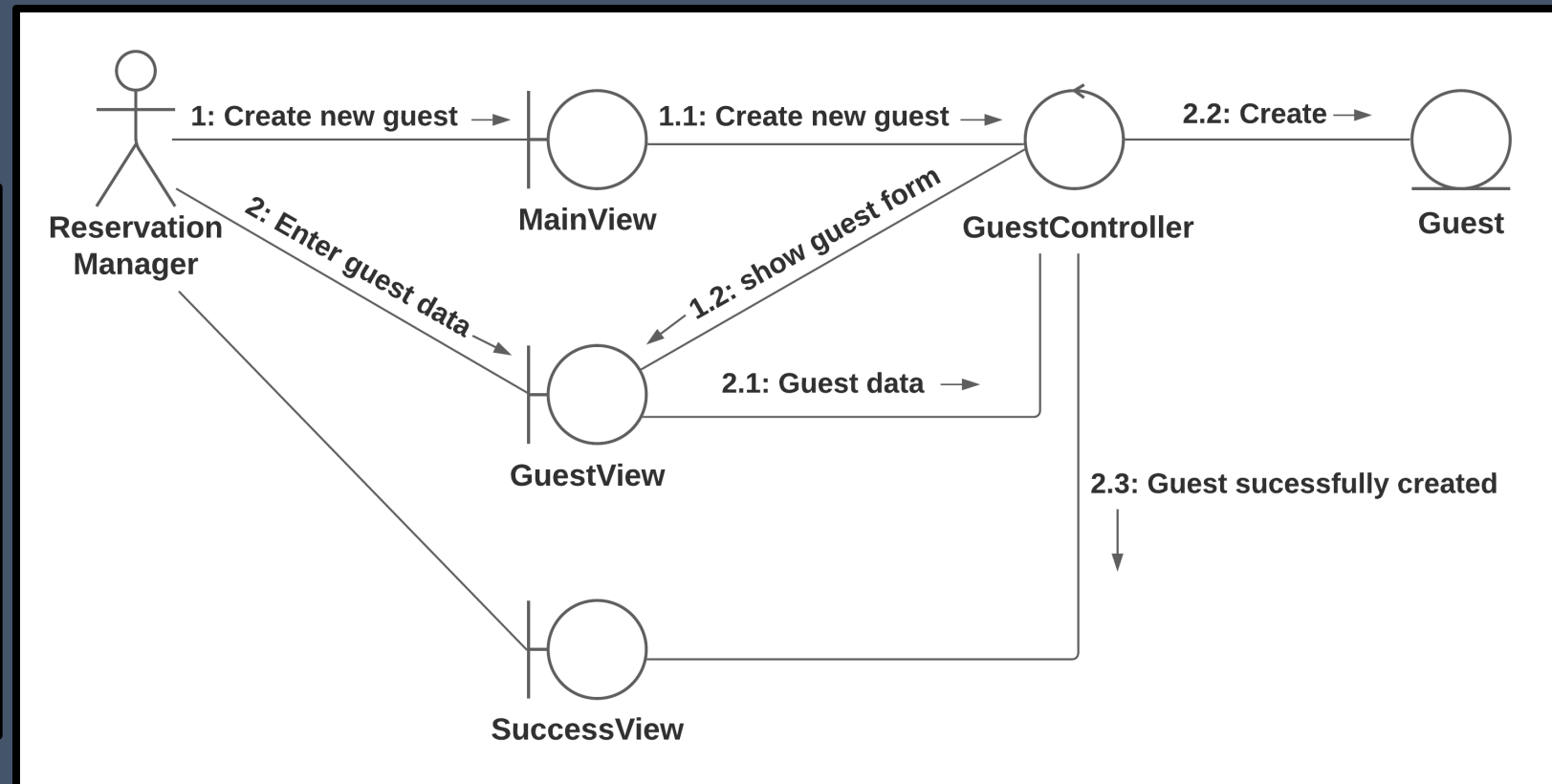
- A screen with an option "Create new Guest" is visible.
- The Reservation Manager is authenticated.

Main flow:

1. The Reservation Manager selects the option "Create new Guest".
2. The system shows a form requesting Guest data.
3. The Reservation Manager enters the form data.
4. The systems shows a message "Guest created sucessfully".

Alternate flow(s):

....
....



Step 3c – Label the actions performed by each component

“Anyone reviewing a robustness diagram should be able to read a course of action in the use case scenario, trace his finger along with the associations on the diagram, and see a clear match between text and picture”.

Use case scenario: Create a Guest

Pre-condition(s):

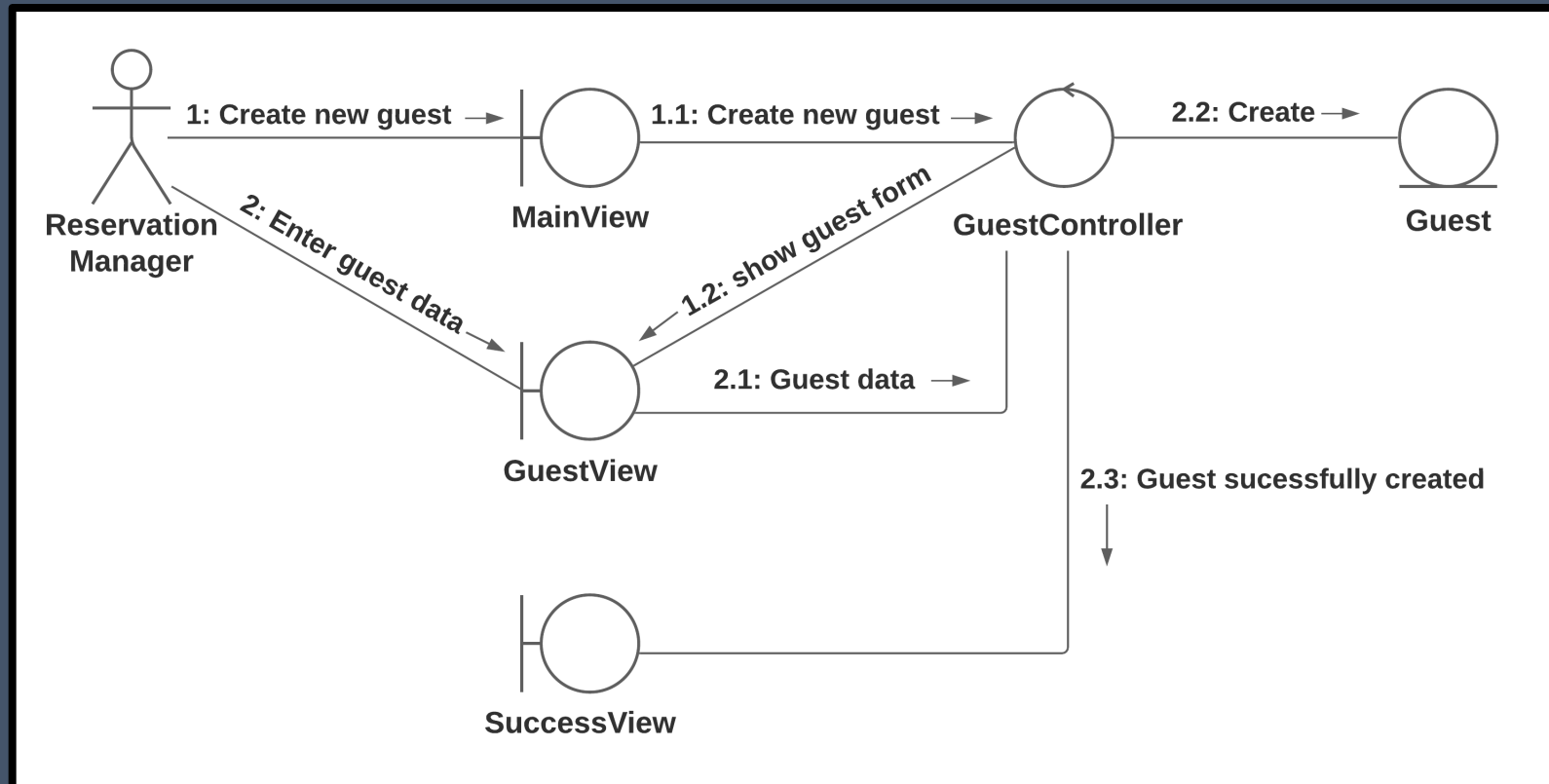
- A screen with an option “Create new Guest” is visible.
- The Reservation Manager is authenticated.

Main flow:

- 1.The Reservation Manager selects the option “Create new Guest”.
- 2.The system shows a form requesting Guest data.
- 3.The Reservation Manager enters the form data.
- 4.The systems shows a message “Guest created sucessfully”.

Alternate flow(s):

....
....



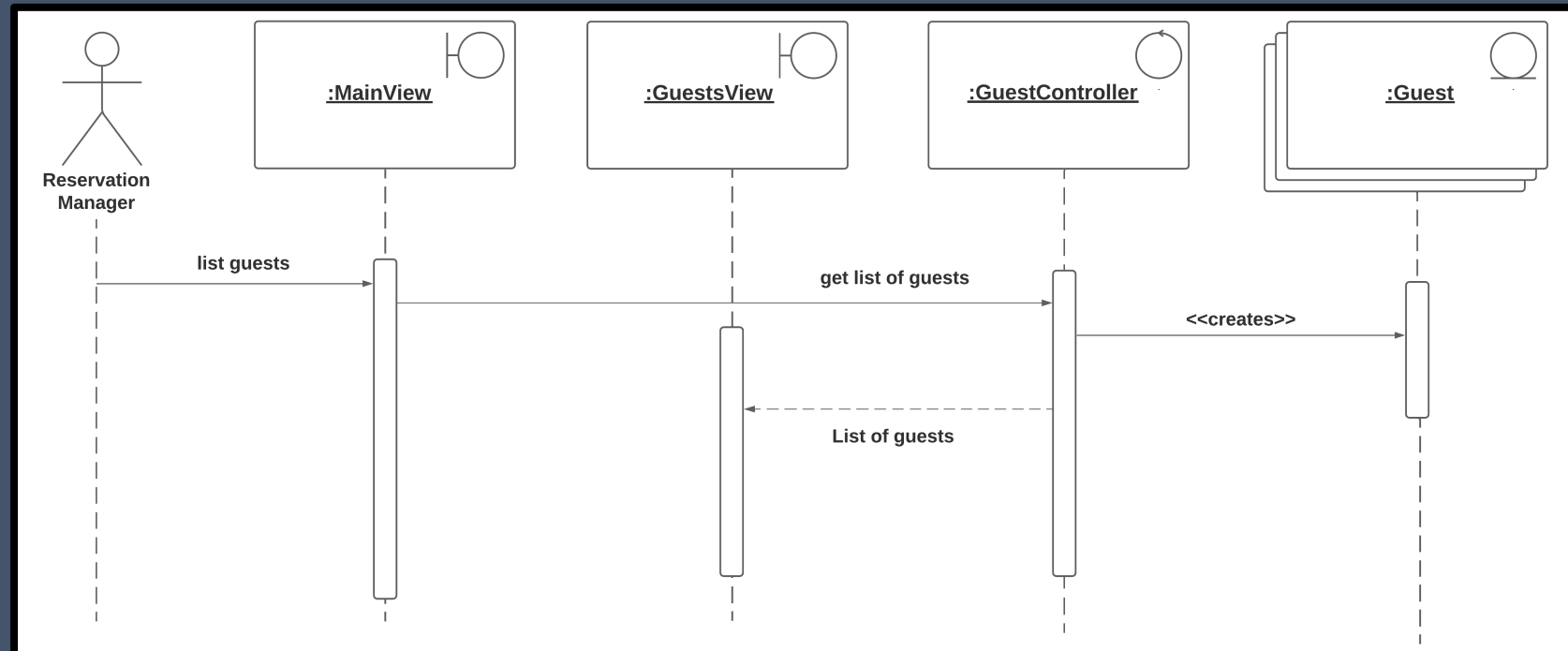
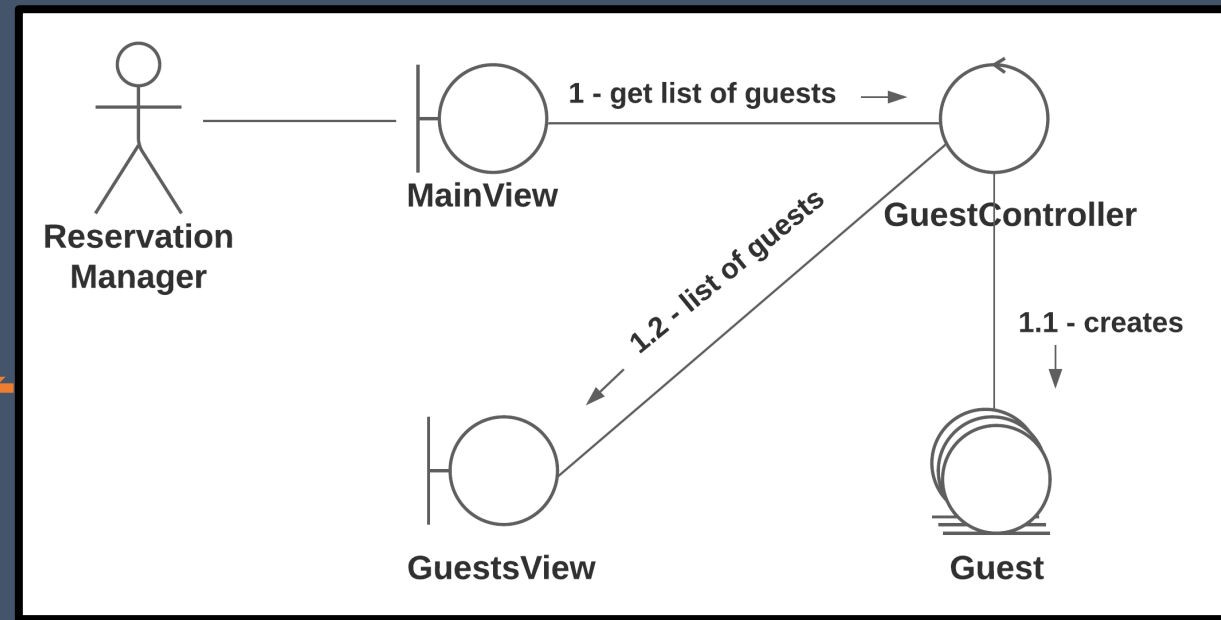
From UML Collaboration diagram to UML Sequence diagram

Lecture: From UML Collaboration diagram to UML Sequence diagram

Module: From requirements model to design model

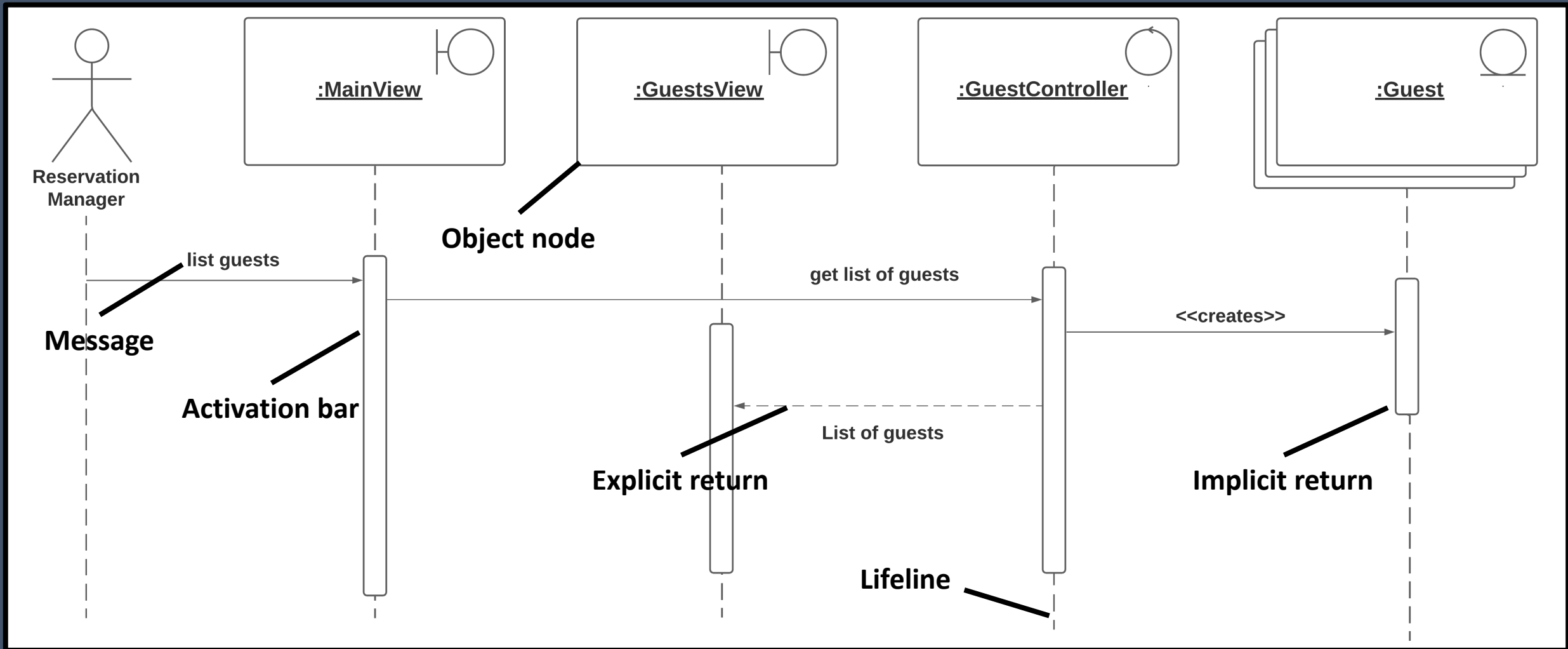
From Collaboration to a Sequence Diagram

- To provide a different perspective on the Robustness model you can convert the Collaboration diagram into Sequence diagram.



UML Sequence diagram

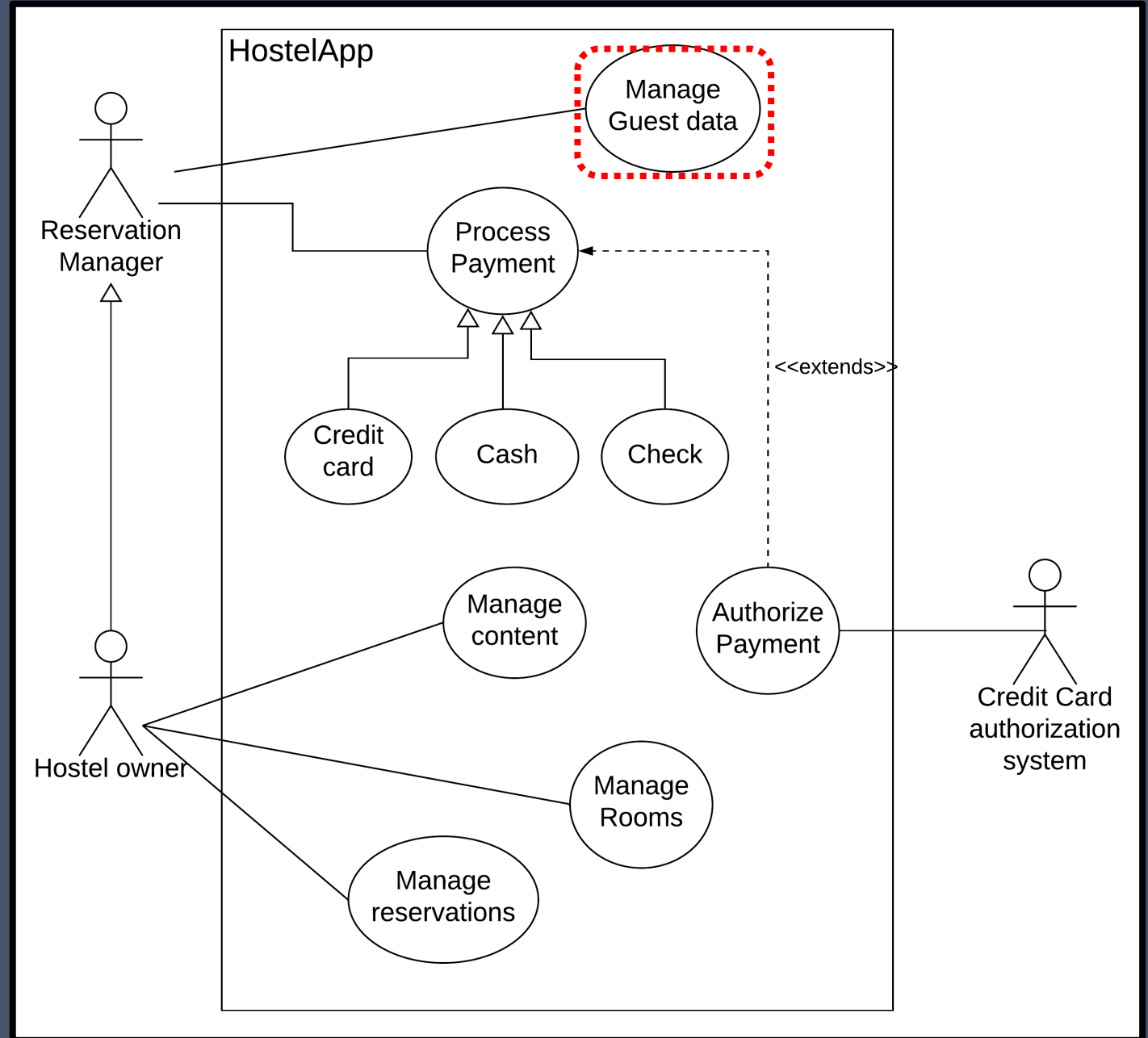
Elements of a UML Sequence diagram



The Design Model with a Sequence diagram

1. Arrange the **design components** at the **top** of the **Sequence diagram** to **reflect the time-order** of the **first activity** that begins with “*the actor ...*” in the **use case scenario**.
2. Add **message links** and **activation bars** for each **message** in the **first activity**.
3. **Repeat Step 2** for each **activity** in the **use case scenario** until the **conversion is complete**.

Use case example



Example #1: List all guests

Use case scenario: List all guests

Pre-condition(s):

- A homepage of the hostelApp with the option "*List all Guests*" is visible.
- The Reservation Manager is authenticated.

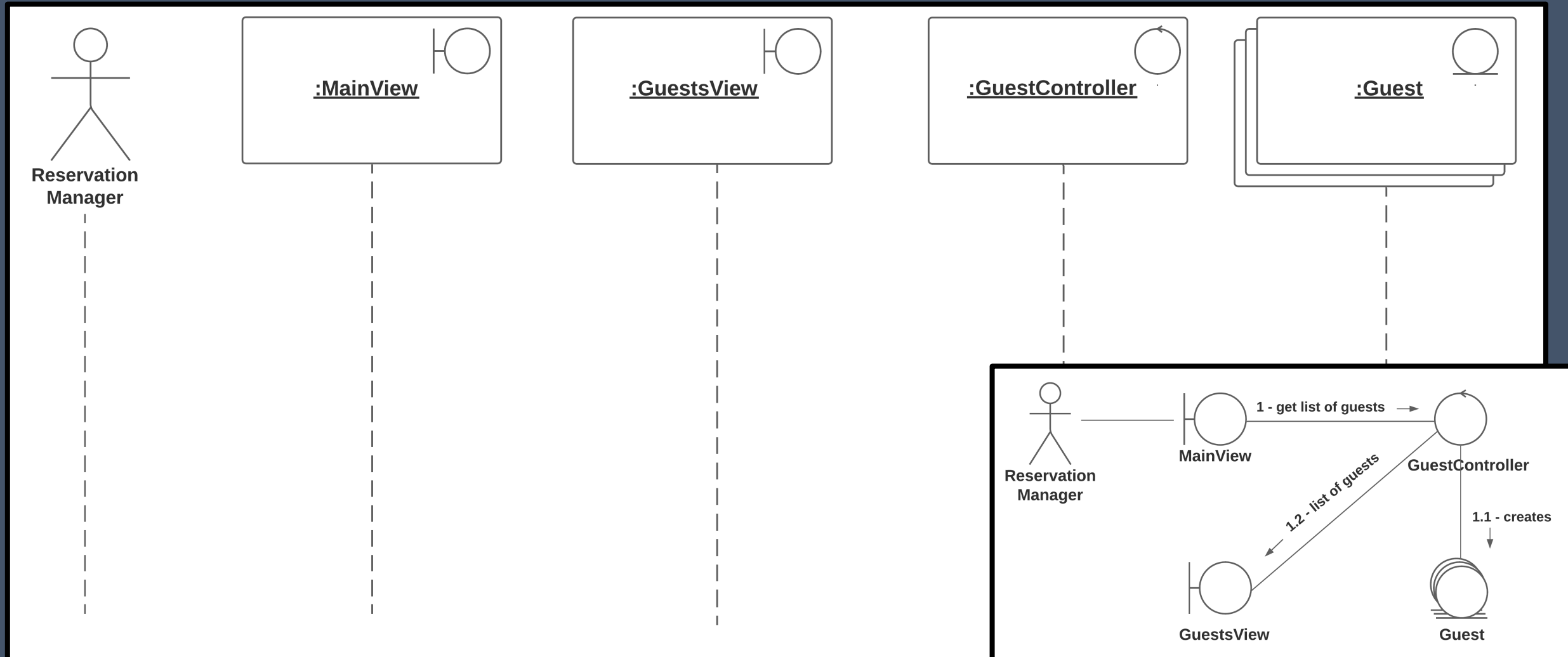
Main flow:

- 1.The Reservation Manager selects the option "*List all Guests*".
- 2.The system shows a page with data of all guests who have already been to the hostel.

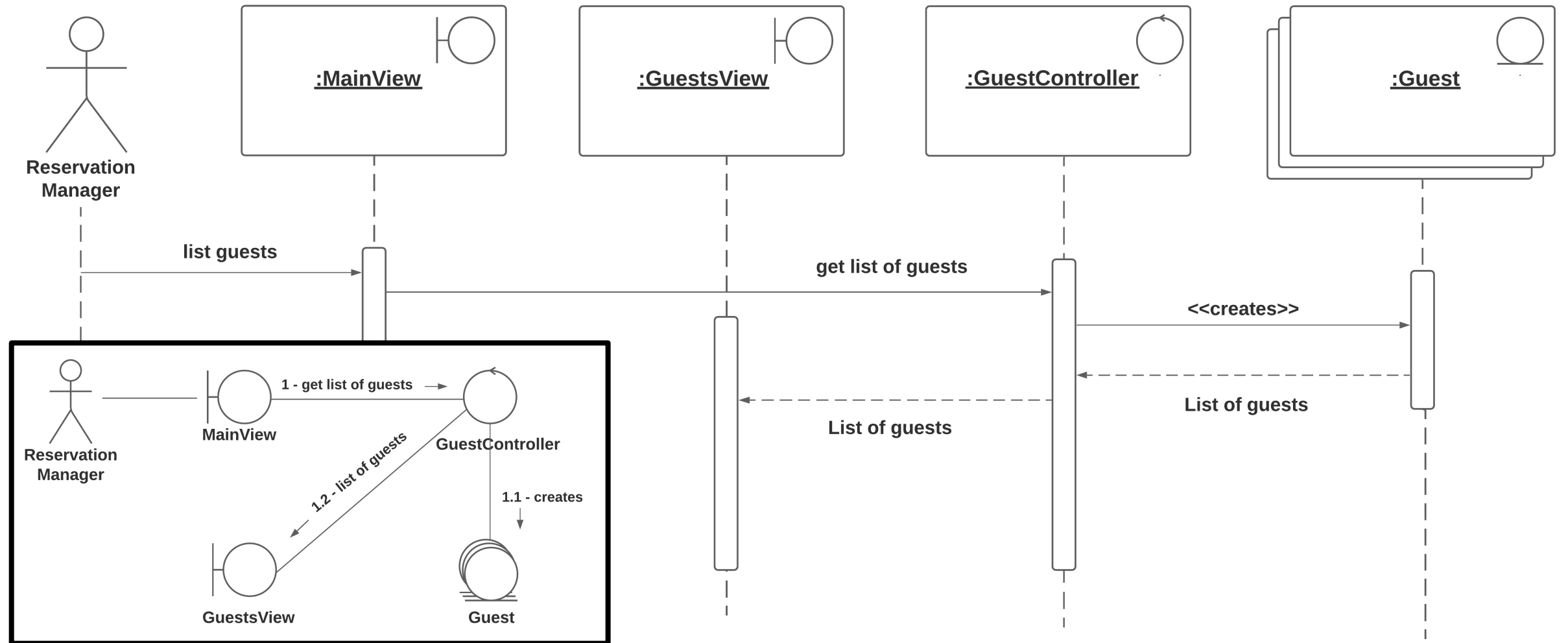
Alternate flow(s):

....
....

1 – Arrange the design components



2 – Add Message Links and Activation Bars



3 – Repeat step 2 for each activity

- Since there are **no more activities** (*the actor...*), we are **done**.

Use case scenario: List all guests

Pre-condition(s):

- A homepage of the hostelApp with the option "*List all Guests*" is visible.
- The Reservation Manager is authenticated.

Main flow:

- 1.The Reservation Manager selects the option "*List all Guests*".
- 2.The system shows a page with data of all guests who have already been to the hostel.

Alternate flow(s):

....
....

Example #2: Create a guest

Use case scenario: Create a Guest

Pre-condition(s):

- A screen with an option "*Create new Guest*" is visible.
- The Reservation Manager is authenticated.

Main flow:

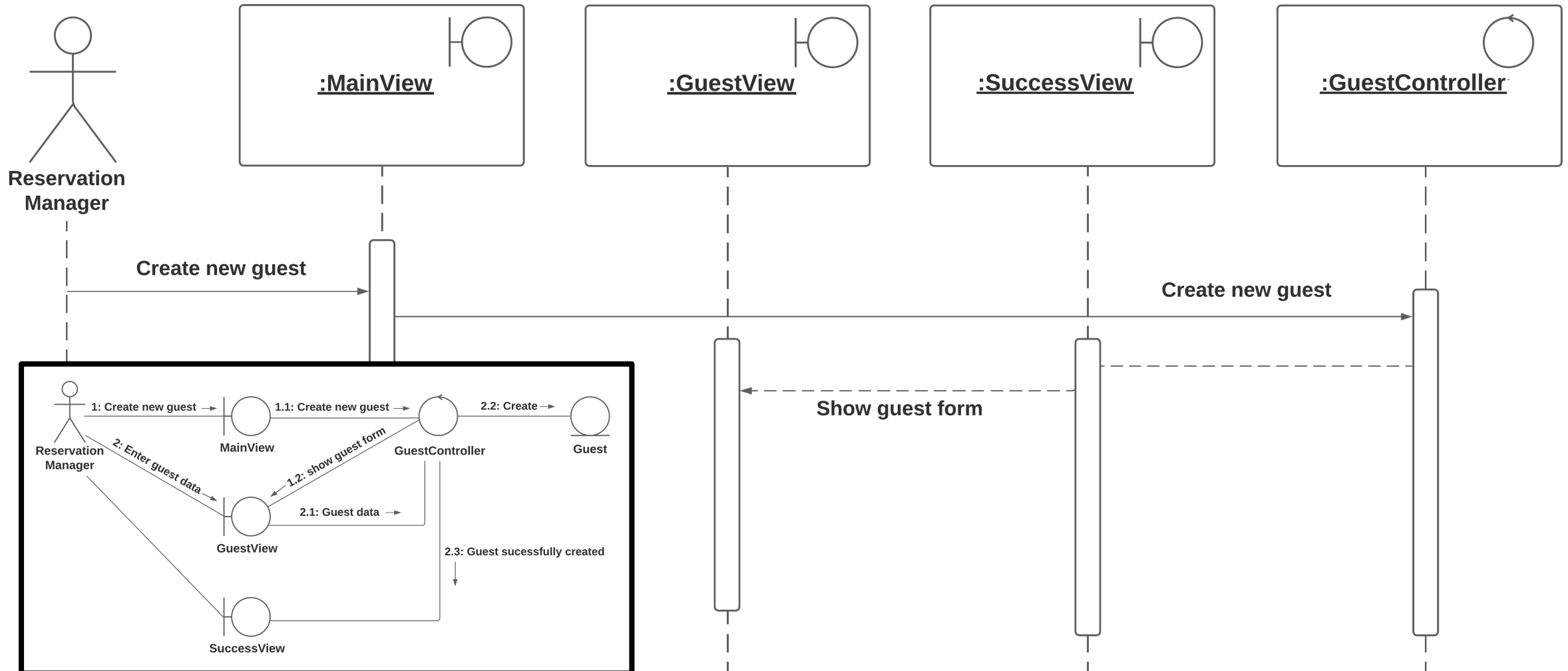
- 1.The Reservation Manager selects the option "*Create new Guest*".
- 2.The system shows a form requesting Guest data.
- 3.The Reservation Manager enters the form data.
- 4.The systems shows a message "*Guest created sucessfully*".

Alternate flow(s):

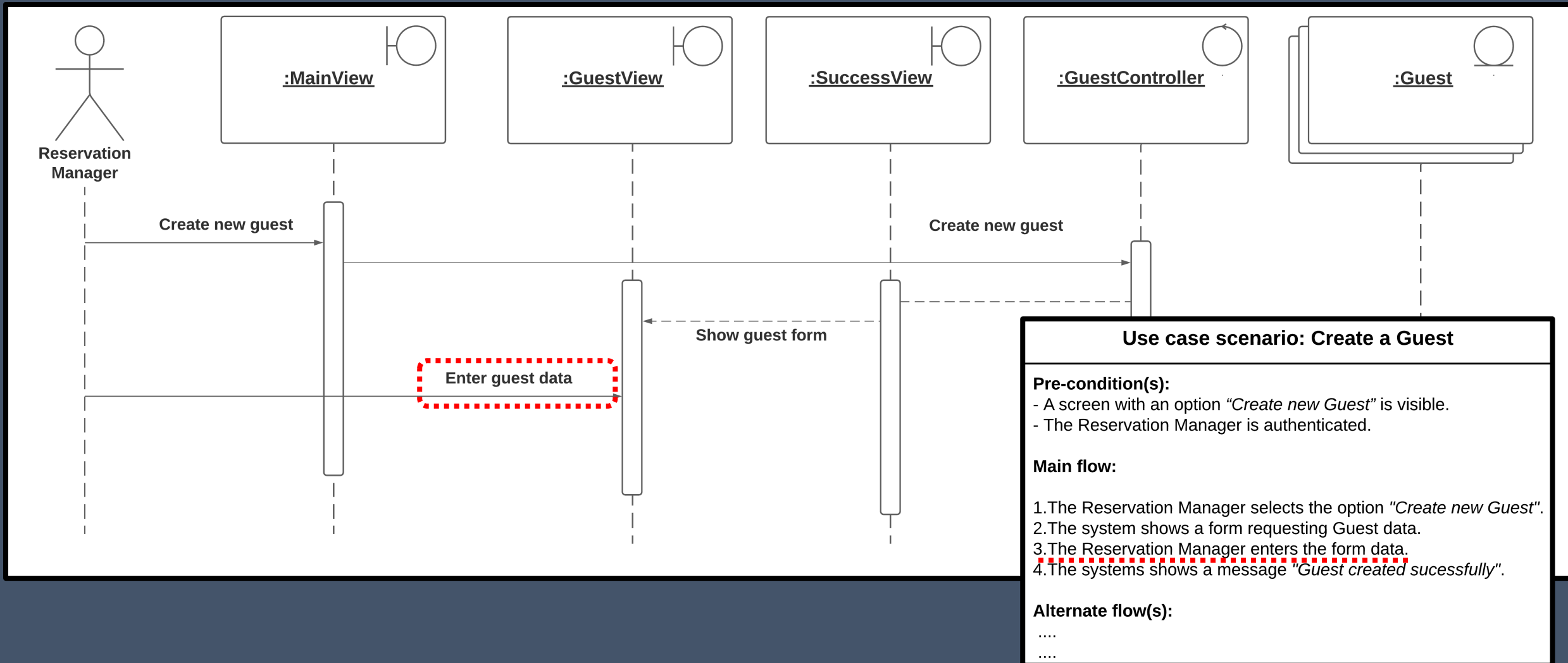
....

....

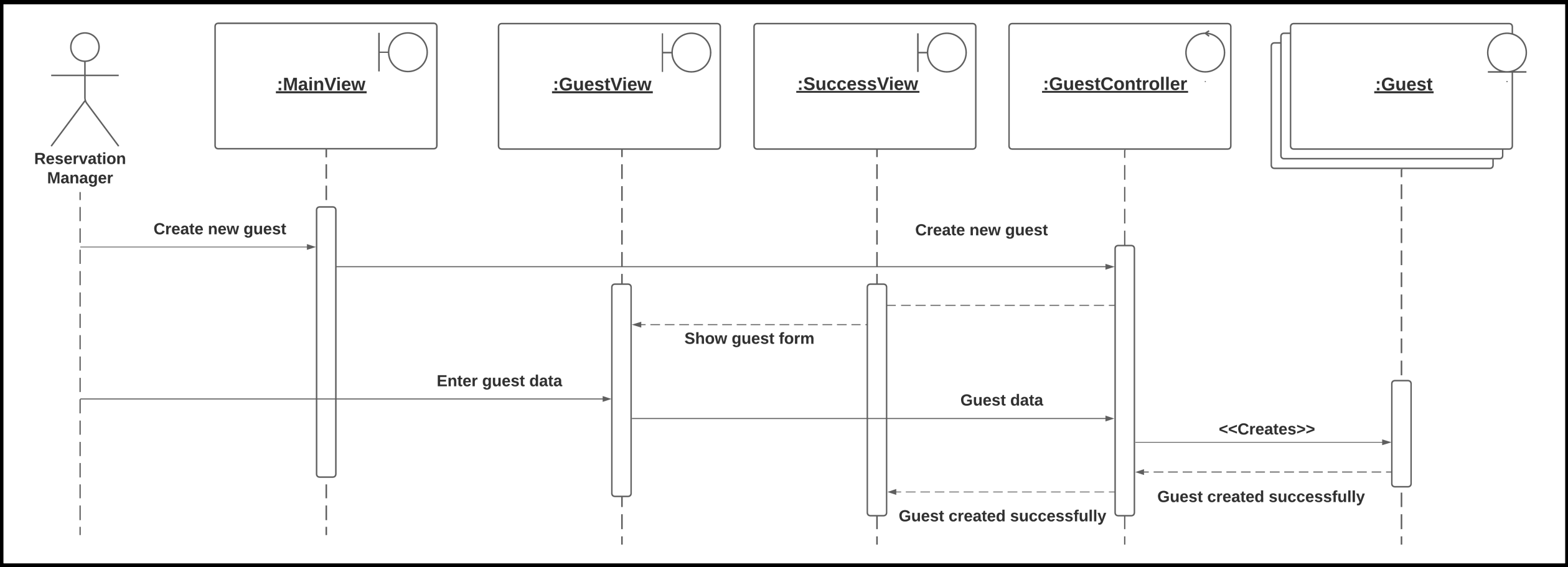
2 – Add Message Links and Activation Bars



3 – Repeat step 2 for each activity



Finished conversion



From Requirements model to Architecture model

Selecting an architecture type

Lecture: Selecting an architecture type

Module: From requirements model to architecture model

Selecting the architecture type

- The **architecture you use depends on many factors**, including:
 - ✓ The **platform constraints** in the **system requirements**.
 - ✓ The **modes of user interaction**.
 - ✓ The **persistence mechanism**.
 - ✓ **Data and transactional integrity**.
- There are **hundreds of successful software architectures**.

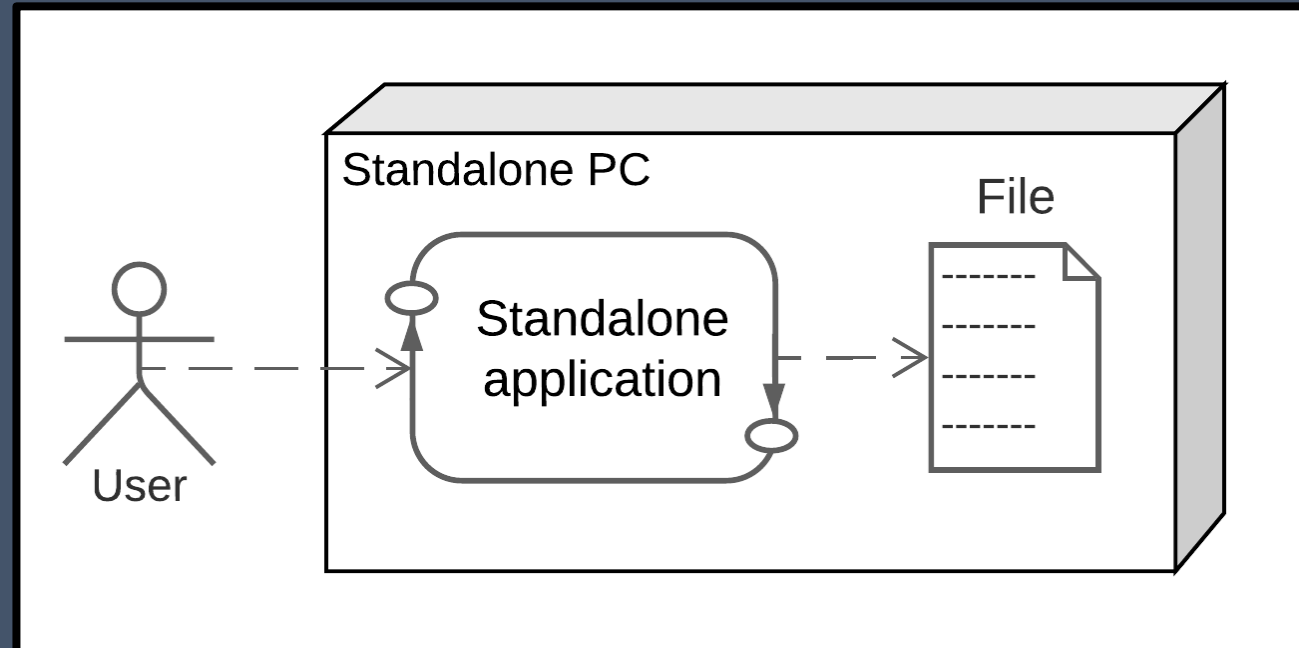
Examples of software architectures

- **Standalone** software architecture
- **2-tiers** software architecture
 - ✓ 2-tiers software architecture (**thick client**)
 - ✓ 2-tiers software architecture (**thin client**)
 - ✓ 2-tiers software architecture (**thin client, web centric**)
- **N-tiers** software architecture
 - ✓ N-tiers software architecture (**thin client**)
 - ✓ N-tiers software architecture (**thin client, web centric**)
 - ✓ N-tiers enterprise software architecture.

Stand-alone software architecture

Standalone software architecture

- **No external data sources.**
 - ✓ All application data exists on a file server.
- **No network communication.**
 - ✓ All application components exist on one machine.



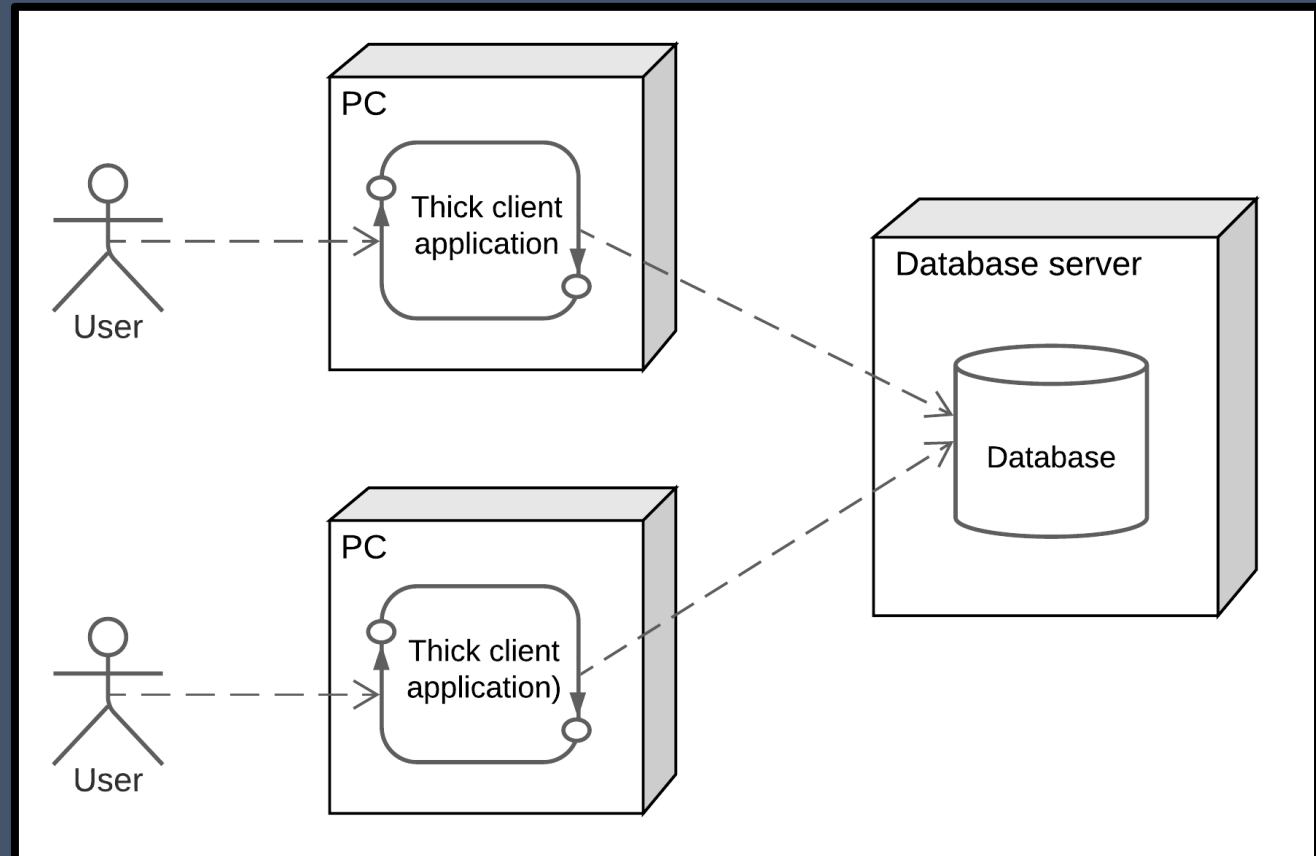
2-tiers software architecture

2-tiers software architecture

- **Three common variations:**
 - ✓ 2-tiers software architecture (**thick client**)
 - ✓ 2-tiers software architecture (**thin client**)
 - ✓ 2-tiers software architecture (**thin client, web centric**)

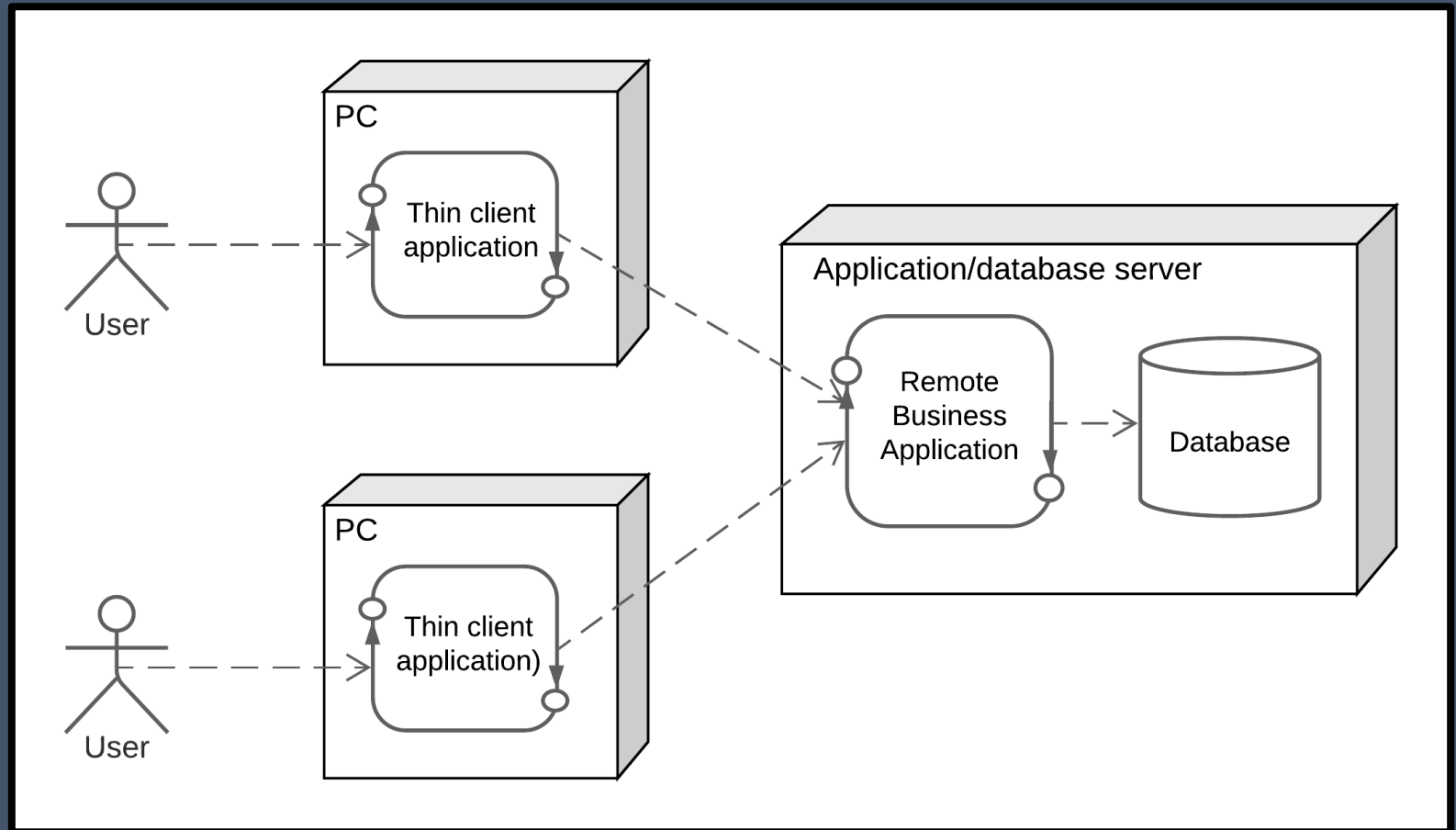
2-tiers software architecture (thick client)

- Thick client
 - ✓ Business logic in the client tier.
- Database manages data integrity.



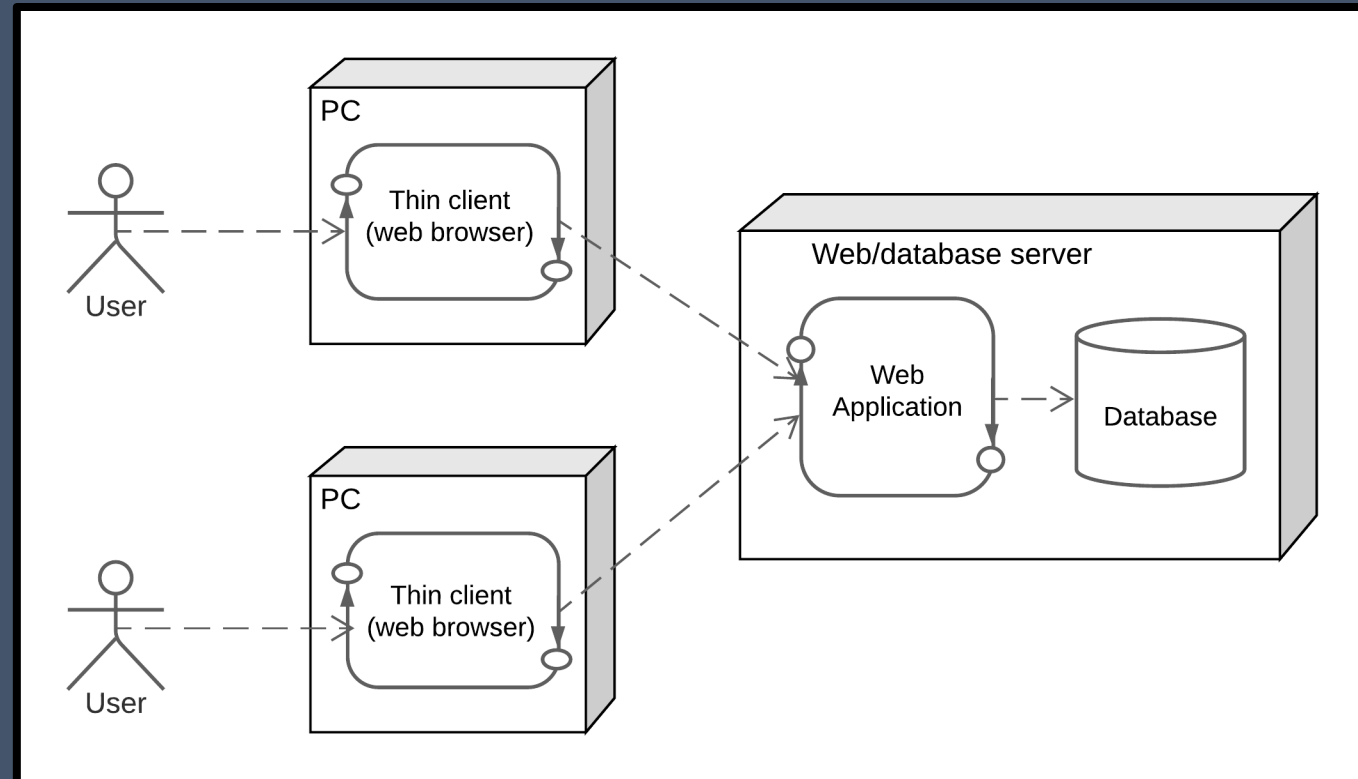
2-tiers software architecture (thin client)

- Thin client
 - ✓ business logic application in the application server.
- One server machine with business application and database.



2-tiers software architecture (web centric)

- **Web browser** becomes the thin client.
- **Web server** provides presentation and business logic.



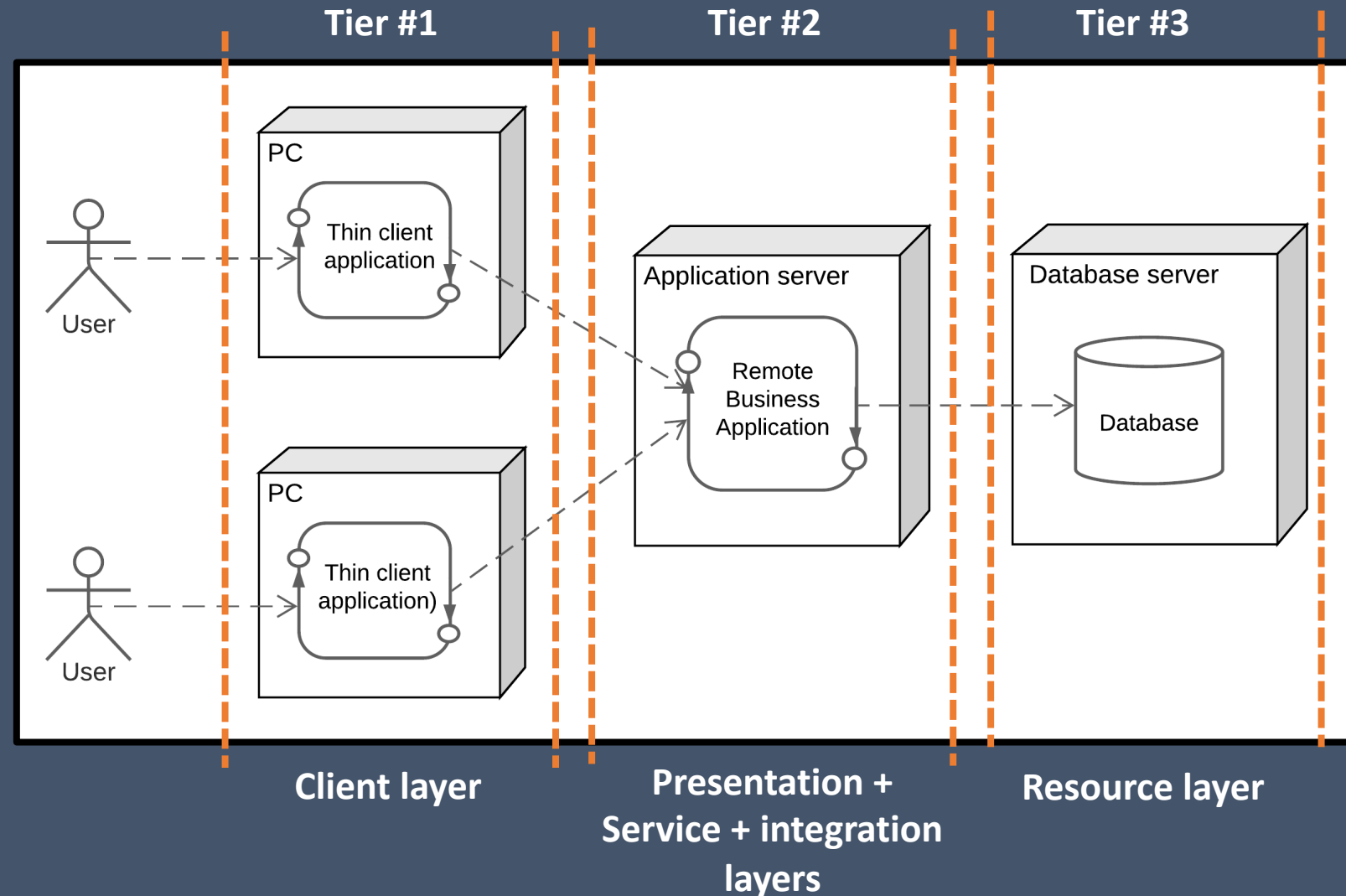
N-tiers software architecture

N-tiers software architecture

- **Three common variations:**
 - ✓ N-tiers software architecture (**thin client**)
 - ✓ N-tiers software architecture (**thin client, web centric**)
 - ✓ N-tiers enterprise software architecture.

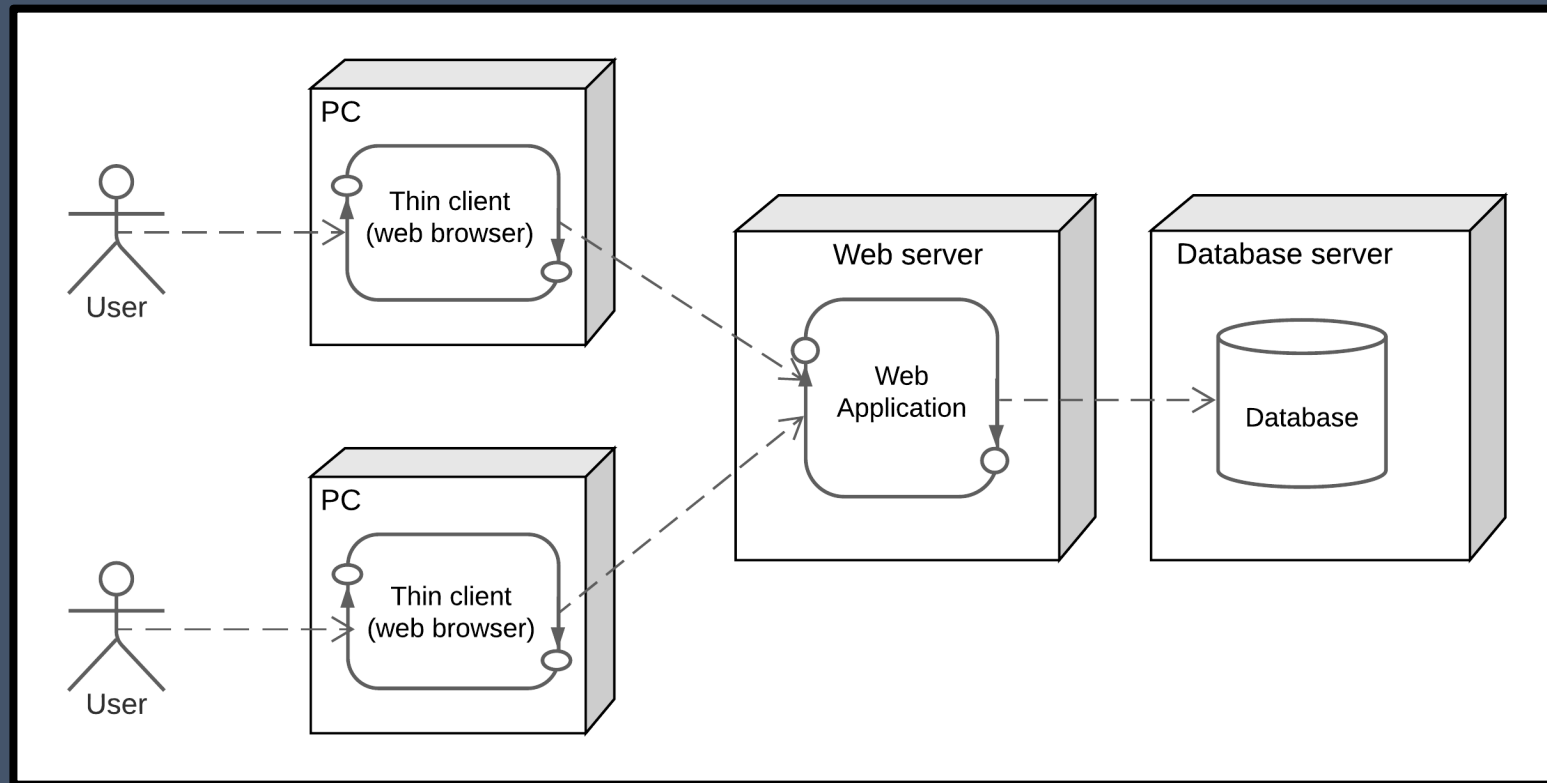
N-tiers software architecture

- Thin client
 - Business logic is in the application server.
- Application server manages data integrity.

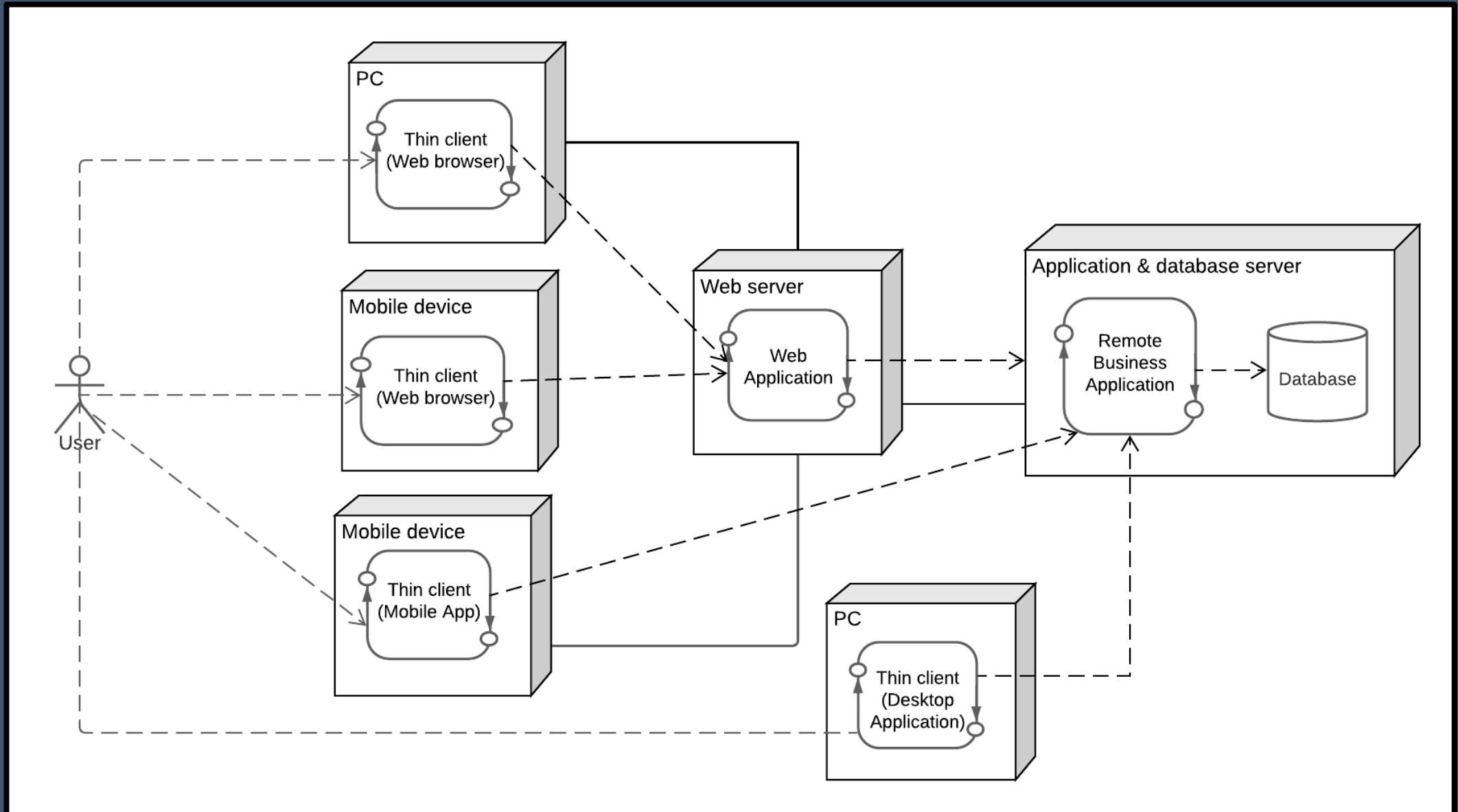


N-tiers web-centric software architecture

- Web browser becomes the thin client.
- Web server provides presentation and business logic.

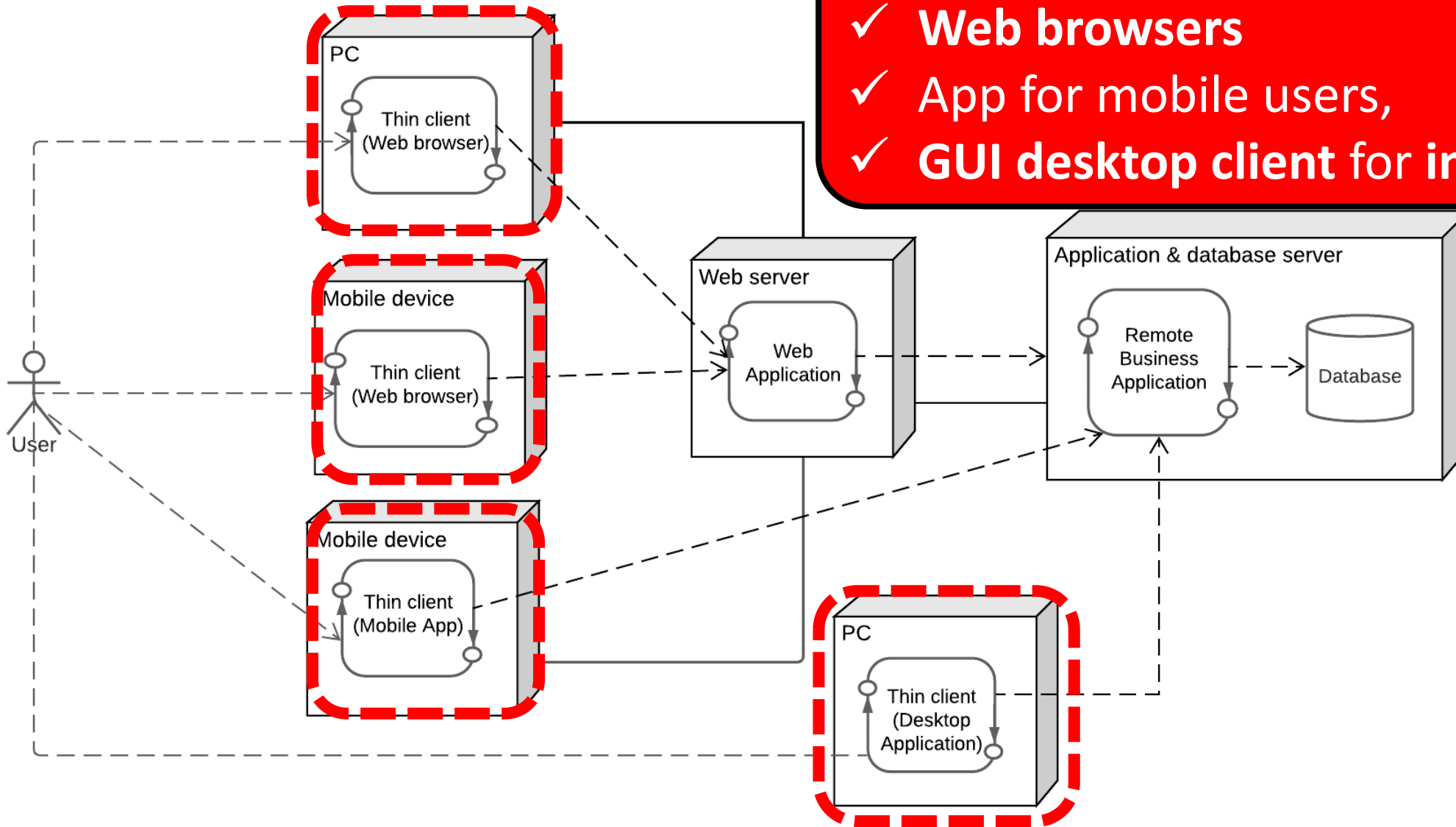


N-tiers enterprise software architecture



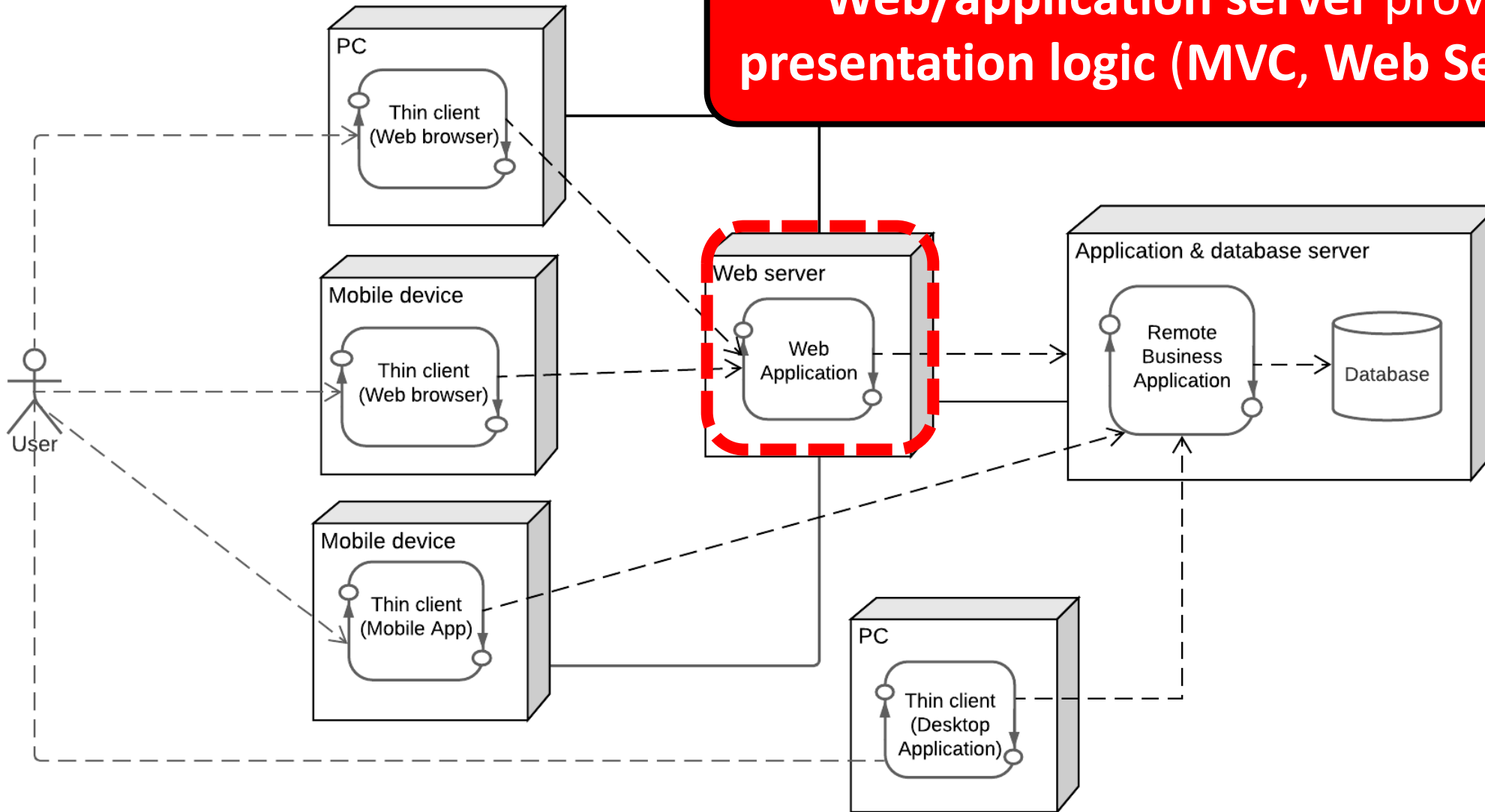
4 thin clients:

- ✓ Web browsers
- ✓ App for mobile users,
- ✓ GUI desktop client for intranet users.



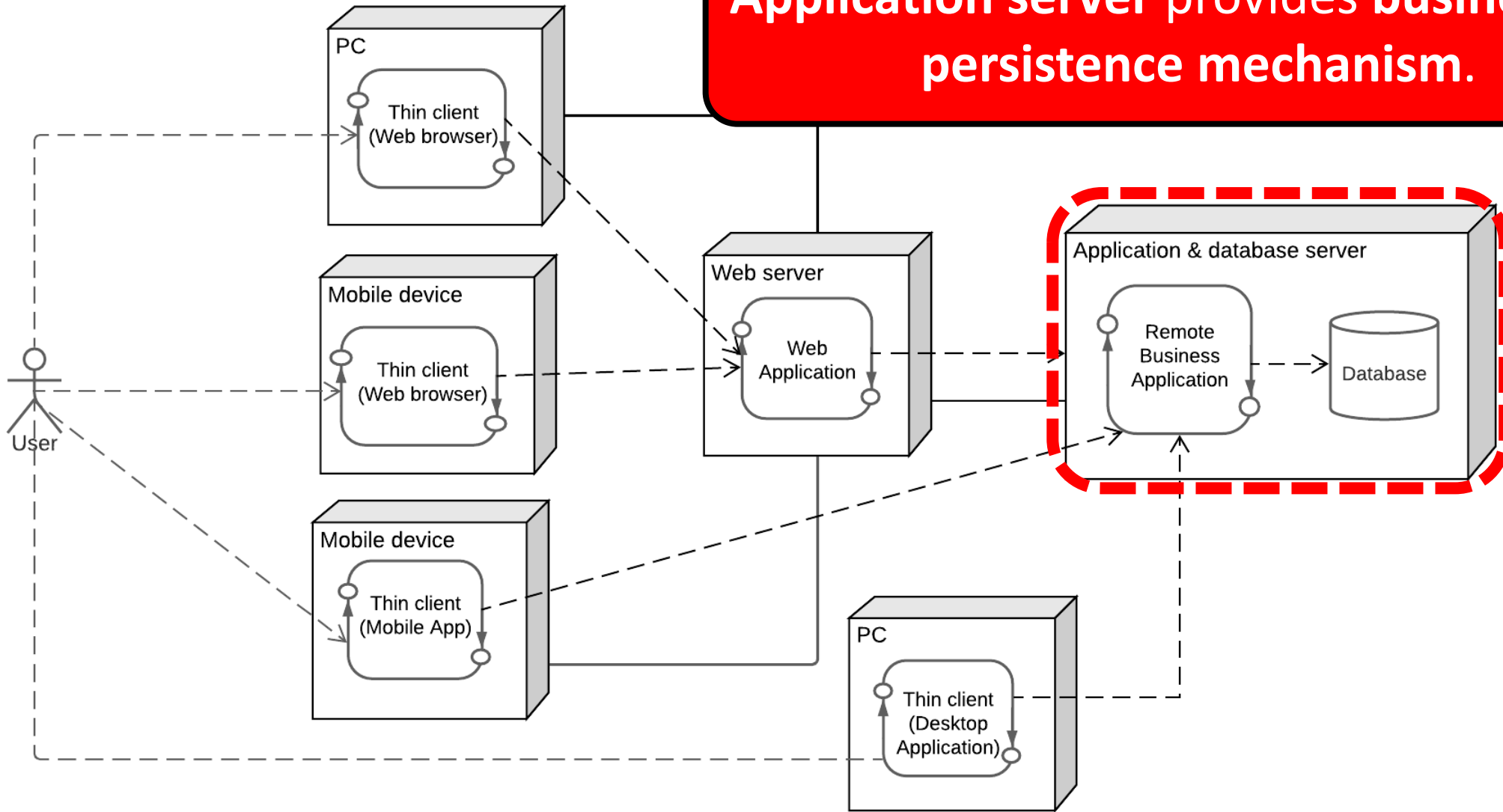
Enterprise (n-tier) software architecture

Web/application server provides presentation logic (MVC, Web Services).



Enterprise (n-tier) software architecture

Application server provides business and persistence mechanism.



Enterprise (n-tier) software architecture

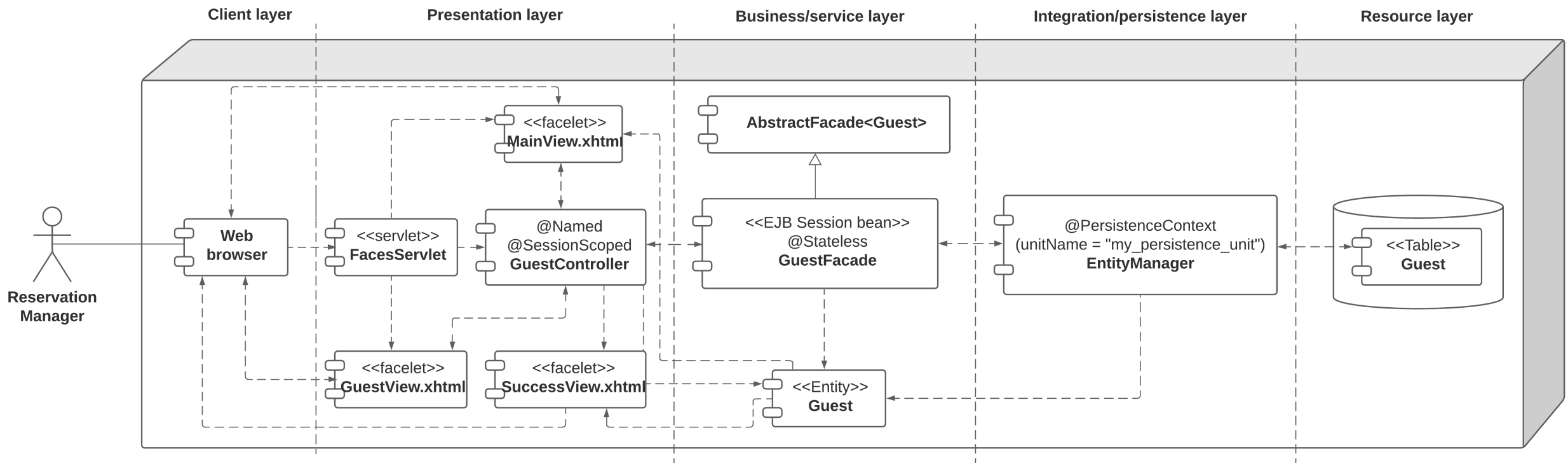
Create the architecture model

Lecture: Detailed Deployment diagram

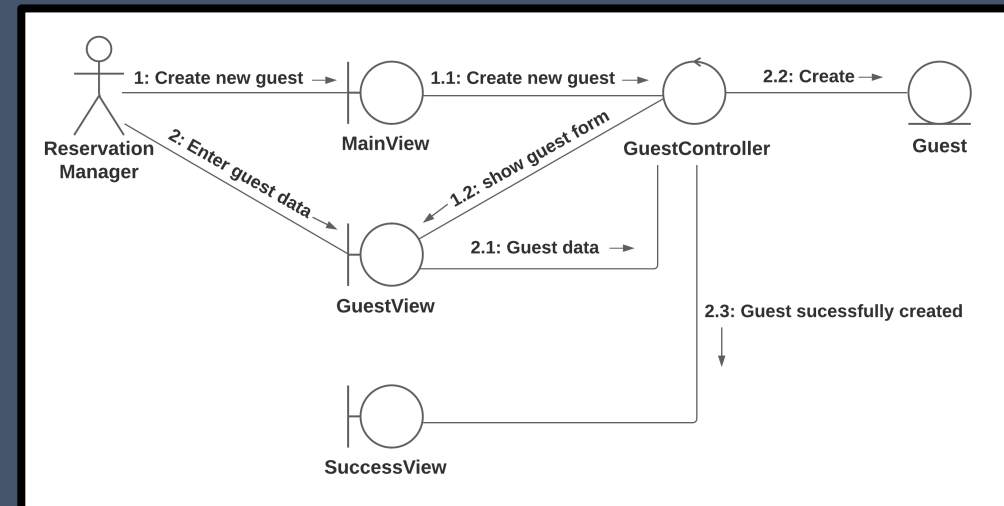
Module: From requirements model to architecture model

Creating the architecture model

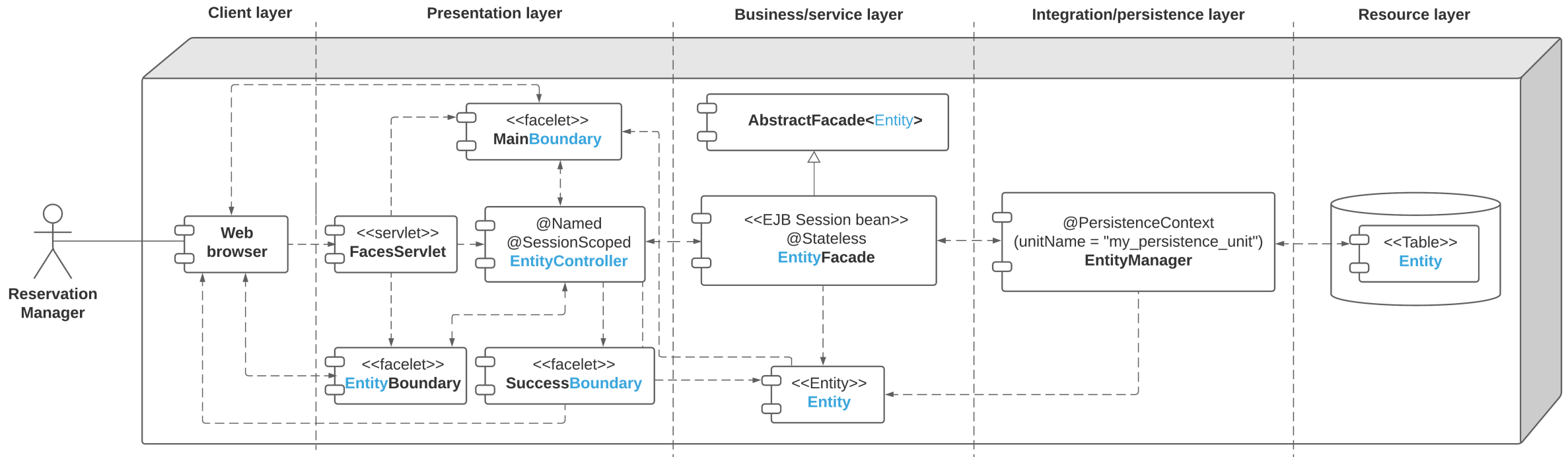
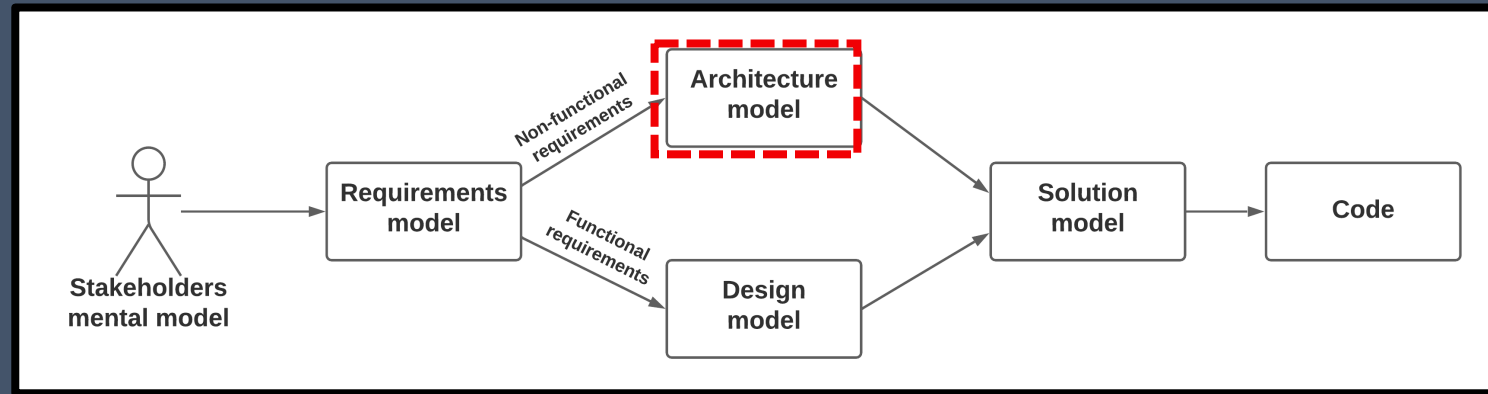
1. **Design the components** for the **architecturally significant use cases**.
2. Place **design components** into the selected **architecture type**.
3. Draw the **architecture model** in a **UML Deployment diagram** by merging the **design** and **architecture components**.



Architecture model



Template of the architecture model



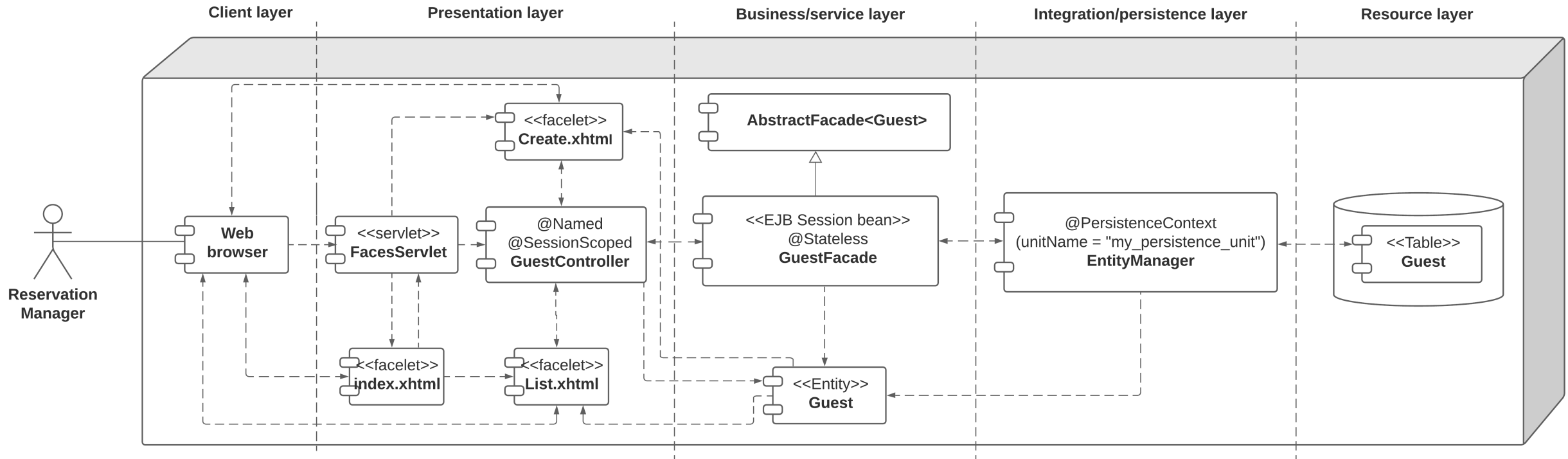
**From Design and Architecture
models to the Solution model**

Create the Solution model

Lecture: Create the Solution model

Module: From Design and Architecture models to the Solution model

Solution model



Questions?